extend Interoperability Suite 11.0.0 Release Notes

Notices

Copyright

Trademarks

Rocket is a registered trademark of Rocket Software, Inc. For a list of Rocket registered trademarks go to: www.rocketsoftware.com/about/legal. All other products or services mentioned in this document may be covered by the trademarks, service marks, or product names of their respective owners.

Examples

This information might contain examples of data and reports. The examples include the names of individuals, companies, brands, and products. All of these names are fictitious and any similarity to the names and addresses used by an actual business enterprise is entirely coincidental.

License agreement

This software and the associated documentation are proprietary and confidential to Rocket Software, Inc. or its affiliates, are furnished under license, and may be used and copied only in accordance with the terms of such license.

Note: This product may contain encryption technology. Many countries prohibit or restrict the use, import, or export of encryption technologies, and current use, import, and export regulations should be followed when exporting this product.

Corporate information

Rocket Software, Inc. develops enterprise infrastructure products in four key areas: storage, networks, and compliance; database servers and tools; business information and analytics; and application development, integration, and modernization.

Website: www.rocketsoftware.com

Rocket Global Headquarters

77 4th Avenue, Suite 100

Waltham, MA 02451-1468

USA

To contact Rocket Software by telephone for any reason, including obtaining pre-sales information and technical support, use one of the following telephone numbers.

Country and Toll-free telephone number

- United States: 1-855-577-4323
- Australia: 1-800-823-405
- Belgium: 0800-266-65
- Canada: 1-855-577-4323
- China: 400-120-9242
- France: 08-05-08-05-62
- Germany: 0800-180-0882
- Italy: 800-878-295
- Japan: 0800-170-5464
- Netherlands: 0-800-022-2961
- New Zealand: 0800-003210
- South Africa: 0-800-980-818
- United Kingdom: 0800-520-0439

Contacting Technical Support

The Rocket Community is the primary method of obtaining support. If you have current support and maintenance agreements with Rocket Software, you can access the Rocket Community and report a problem, download an update, or read answers to FAQs. To log in to the Rocket Community or to request a Rocket Community account, go to www.rocketsoftware.com/support. In addition to using the Rocket Community to obtain support, you can use one of the telephone numbers that are listed above or send an email to support@rocketsoftware.com.

Table of contents

extend Interoperability Suite 11.0.0 Release Notes **5**
- extend System Requirements 5
- Installation Notes 10
- New Features in extend 11.0.0 14
- Known Issues 18
- Enhancements 18
- Resolved Issues 20
- Contacting Rocket Software 28
- Trademark and Legal Information 29

extend Interoperability Suite 11.0.0 Release Notes

These release notes contain information that might not appear in the Help. Read them in their entirety before you install the product.

- ***extend*** System Requirements
- Installation Notes
- New Features
- Known Issues
- Enhancements
- Resolved Issues

extend System Requirements

This topic provides information about hardware, operating system, database, and other requirements for running ***extend*** products.

- Hardware Requirements
- Supported Operating Systems
- Supported Relational Database Management Systems (RDBMSs)
- extend Web Help
- Additional Requirements

Hardware Requirements

extend software has the following requirements:

For Windows:

- The amount of disk space needed to install the ACUCOBOL-GT development system is typically less than 35 MB.
- The production system requires at least 12 MB for installation.
- You need an additional 14 MB to install the RDBMS products.
- Use of .NET controls with the runtime and thin client requires .NET Framework 4.8.

For all other platforms:

The amount of disk space needed to install all ***extend*** products requires at least 105 MB.

Supported Operating Systems

All supported operating systems are both 32- and 64-bit unless otherwise noted.

		Rocket Software Product Release			
Operating System Base	**Processor**	**10.4.1**	**10.5.0**	**10.5.1**	**11.0**
AIX 6.x	PowerPC	☑	☑		

		Rocket Software Product Release			
Operating System Base	**Processor**	**10.4.1**	**10.5.0**	**10.5.1**	**11.0**
AIX 7.x[1]	PowerPC	☑	☑	☑	☑ [2]
AIX 7.3	PowerPC				☑
FreeBSD 9.x	x86-64[3]	☑			
HP/UX 11v3	Itanium-2[4]	☑	☑	☑	
HP/UX 11v3	PA-RISC	☑			
Linux glibc 2.5 and later[5]	x86-64	☑	☑	☑	☑
Linux glibc 2.5 and later[5]	PowerPC	☑	☑		
Solaris 10	SPARC[4]	☑	☑	☑	
Solaris 11.3	Intel				☑
Solaris 11.3	SPARC	☑	☑	☑	☑
Windows XP	x86-64[3]	☑			
Windows Server 2003	x86-64[3]	☑			
Windows 7	x86-64	☑	☑	☑	
Windows 8.1	x86-64	☑	☑	☑	
Windows 10	x86-64	☑	☑	☑	☑
Windows 11	x86-64	☑	☑	☑	☑
Windows Server 2008 R2 SP1	x86-64	☑	☑	☑	
Windows Server 2012 R2	x86-64	☑	☑	☑	
Windows Server 2016	x86-64	☑	☑	☑	
Windows Server 2019	x86-64	☑	☑	☑	☑
Windows Server 2022	x86-64		☑	☑	☑
Windows Server 2025	x86-64				☑

		Rocket Software Product Release			
Operating System Base	**Processor**	**10.4.1**	**10.5.0**	**10.5.1**	**11.0**

[1] For the AIX 7.1 platform, the minimum requirement is version 7.1 Technology Level 4 (7100-04)).

[2] Minimum requirement is AIX 7.2 SP1.

[3] 32-bit only.

[4] Support for this platform ends after ***extend*** version10.5.1.

[5] For AcuToWeb, use Linux glibc 2.7 and later.

Supported Relational Database Management Systems (RDBMSs)

extend supports the base version and all incremental updates to that base version.

	Rocket Software extend Product Release				
RDBMS base	**10.0**	**10.3**	**10.4**	**10.5**	**11.0**
Oracle 11	☑	☑	☑	☑	☑
Oracle 19		☑	☑	☑	☑
Oracle 19c					☑
Oracle 21c		☑	☑		
Oracle 23c					☑
Microsoft SQL Server 2012	☑	☑	☑	☑	☑
Microsoft SQL Server 2016	☑	☑	☑	☑	☑
Microsoft SQL Server 2017					☑
Microsoft SQL Server 2019			☑	☑	☑
Microsoft SQL Server 2022		☑		☑	☑
IBM DB2 LUW 10	☑	☑	☑	☑	☑
IBM DB2 LUW 10.5			☑	☑	☑
IBM DB2 LUW 11		☑	☑	☑	☑
IBM DB2 LUW 11.1					☑
IBM DB2 LUW 11.5					☑
Informix 12			☑	☑	☑

	Rocket Software extend Product Release				
RDBMS base	10.0	10.3	10.4	10.5	11.0
Informix 14			☑	☑	☑

extend Web Help

extend Web Help is supported in most currently available browsers.

Additional Requirements

ACUCOBOL-GT

ACUCOBOL-GT .NET 8 applications require the .NET 8 SDK

AcuServer:

- Each server machine must be networked to UNIX, Linux, or Windows clients with TCP/IP. TCP/IP is not sold or supplied by Rocket Software.
- All servers must have a copy of the AcuServer license management file.
- Windows clients can run any TCP/IP software that uses a `WINSOCK2` compliant `ws2_32.dll`.
- Unless you have an unlimited license for AcuServer, all UNIX servers must run the current version of `acushare`, which is included on the AcuServer distribution media.
- All servers must have a copy of the license file activated by the product installation script. This file is named `acuserve.alc`.
- Client machines must have an ACUCOBOL-GT AcuServer-enabled runtime. All Windows runtimes Version 5.0 and later are AcuServer-enabled. To verify that your UNIX runtime is AcuServer-enabled, type `runcbl -v` in a Command prompt and look for this line.

  ```
  AcuServer client
  ```

AcuBench:

- Intel Pentium III CPU, 300 MHz; Intel Pentium IV, 2 GHz recommended
- 128 MB of RAM recommended
- 120 MB of available hard disk space recommended
- mouse
- 800 x 600 VGA display or better; 1024 x 768 VGA display recommended

AcuToWeb:

Java

AcuToWeb requires Java 1.8 or later.

> **Troubleshooting:** We recommend that you install Java using its `.msi` installer rather than installing manually. Manually installed Java is known to cause the following error to appear when starting the **AcuToWeb Control Panel**:

```
Java 1.8 or higher is required for AcuToWeb
```

If you encounter this message, reinstall Java using the `.msi` installer. For example, reinstall using the `.msi` installer for Azul OpenJDK. When finished, try starting the **AcuToWeb Control Panel** again.

Font

Install the MS Sans Sarif font to use with AcuToWeb.

Supported Web Browsers

AcuToWeb supports following Web browsers, and has been tested with the versions indicated:

Browser	Supported Versions
Google Chrome	129 and later
Microsoft Edge	129 and later
Mozilla Firefox	130 and later
Safari	18 or later

AcuSQL:

- Your COBOL application must run on a Windows system or a UNIX system supported by Rocket Software. Unless otherwise indicated, the references to Windows in this manual denote supported Windows operating systems. Where necessary, individual versions of those operating systems are referred to by their specific version numbers.
- AcuSQL must be installed with the ACUCOBOL-GT development system on your Windows or UNIX system.
- If using a database other than Microsoft SQL Server, you must have a working ODBC level 2 API connection to your database, including any required networking software support.
- For SQL Server, if running the AcuSQL interface to Microsoft SQL Server, you must have the SQL Server client software from Microsoft. Use the Query Analyzer to see if the SQL Server client software from Microsoft is on your system. For information on opening the Query Analyzer, see the SQL Server client documentation. If the Query Analyzer opens and you are able to connect to the database, the client libraries are most likely all present. Your SQL Server data source may be hosted on one or more of the supported server operating systems.
- If you are running the AcuSQL interface to MySQL, you must have the following software:
    - MySQL 5.0 Database Server Version 5.0.18 or later (Generally Available release). Testing was done with MySQL 5.0.18 Standard.
    - MySQL Connector/ODBC Version 3.51.11 or later (Generally Available release). Testing was done with the `libmyodbc3-3.51.12.so` library. This file is available from http://dev.mysql.com.

  You can check the version of your server by connecting using `mysql`. The version prints upon connection. For example:

  ```
  [testing ]: mysql
  Welcome to the MySQL monitor. Commands end with ; or \g.
  Your MySQL connection id is 29 to server version:
  5.0.18-standard
  Type 'help;' or '\h' for help. Type '\c' to clear the buffer.
  ```

Once in MySQL, you can also use the following:

```
mysql> select version();
+-----------------+
| version()       |
+-----------------+
| 5.0.18-standard |
+-----------------+
1 row in set (0.09 sec)
```

- If your application accesses DB2 data, IBM's DB2 Connect™ software is recommended. Access to DB2 databases has been tested with DB2 Connect. However, any vendor's properly configured ODBC level 2 API connectivity software should work. Your DB2 data source may be hosted on one (or more) of the supported operating systems.

Acu4GL (for ODBC) driver requirements:

Your ODBC driver must include the following functions:

- all Core ODBC driver functions
- the Level 1 function SQLColumns
- the Level 1 function SQLTables

Depending on the method of record locking you choose, your driver may also need to support some of the following function calls:

- SQLSetStmtOption
- SQLSetScrollOptions
- SQLExtendedFetch
- SQLSetPos

See A_ODBC_LOCK_METHOD in the ***extend*** online help for more information.

To test the capabilities of your ODBC driver, we have included a driver test program on your Acu4GL for ODBC installation disks. You can also consult your driver documentation to ensure that it meets these requirements.

Installation Notes

This section contains information on installing this release.

Back to top

- Before Installing
- Windows Installation
- UNIX Installation

Before Installing

Back to Installation Notes

Download the product

1. Log into the Rocket Customer Community site at https://my.rocketsoftware.com/RocketCommunity/s/.

2. Click **Downloads**.
3. Locate and select the product to download in the **SELECT A PRODUCT** pane.
4. Click on the link in the center pane for the product you want to download.
5. Click the required file link in the **File Name column** on the relevant row.

Extract the installation files

Extract the downloaded product files to a local directory.

If you are upgrading from an earlier release

Clean up your local environment as follows:

- If you have a version of AcuServer, AcuConnect, or AcuToWeb running on its default port, shut it down before proceeding.
- Be sure that you do not have another version of the ***extend*** Interoperability Suite referenced in the PATH system environment variable, as having more than one version specified could cause unexpected results.
- Uninstall any existing ***extend*** products.
- UNIX platforms only - Ensure that the `tar` utility is on your PATH.

Windows Installation

Back to Installation Notes

This section contains steps for installing on Windows platforms.

> **Note:** For additional information regarding Windows installations including licensing, silent installation, modifying your installation, redistribution, and so on, see *Windows Installation* in your ***extend*** *Interoperability Suite* documentation.

> **Important:** You must have a product code and key to activate products included in the ***extend*** Interoperability Suite; however, you can install any or all products now, and activate those products for which you do not have a license at a later date. Be aware that if you attempt to use products for which you do not have a license, you might receive error messages indicating that no license file is available.
>
> In this documentation and in the **Setup Wizard**, the terms *product* and *feature* are equivalent.

1. From the directory where you downloaded the product, run the `.exe` file as an administrator and follow the Wizard instructions to install.

   > **Important:** You must run the `.exe` as an administrator to ensure that you can configure components to start as Windows services where appropriate.

2. On the **Select Installation Folder** page, click **Browse** and select an installation directory or directories. Alternatively, you can accept the default location(s), and then click **Next**.

   > **Restriction:** If you specify a mapped drive, it must map to a local directory; remote mapped drives are not supported.

3. When you get to the **Choose Setup Type** Wizard page, select the type of installation that best meets your needs as described on the page, and then click **Next**.

> **Note:** The **Typical** installation installs the Runtime, AcuConnect, Thin Client, and AcuToWeb. However, we recommend that if you have any plans to use ***extend*** products outside of these products that you either select **Complete** to install all products, or **Custom** to select additional products. You may also rerun the installation Wizard at a later time to add products if you choose.

4. Follow the instructions that correspond to the setup type you have selected:

   Typical

   Continue with step **5**.

   Complete

   If you intend to install BIS or .NET Framework, follow the instructions on this Wizard page. If necessary, click **Cancel** to exit the installation. Otherwise, click **Next** and continue with step **5**.

   Custom

   On the **Product Selection** Wizard page, expand the tree view as necessary, activate the features and sub-features you want to install, and then click **Next**.

   > **Tips:**
   > - The text on the right side of the Wizard provides additional information about selecting features and disk space.
   > - Select **Documentation** on this page to install and access the product user guides from the product UI.

5. On the **Installation Settings** page, ensure that both **Install License Activator** and **Launch License Activator** are both checked.

   The other options on this page are based on the bitism of your installation file (x64 or x86), your chosen Setup Type, and other previous selections when applicable.

6. You may optionally select an alternative bitism for any specific product that is not grayed out on this page.
7. Optionally check **Enable/Disable Start Services buttons**. If you check this box, review the text on the Wizard page and continue as required.
8. Click **Next**.

   If you have opted to install Xcentrisity Business Information Server, the **Logon Information** page appears. In this case, optionally provide the **User Name** and **Password** as described on the page, and then click **Next**.

9. Click **Install**.

   Toward the end of the installation process, the **Activator Wizard** appears.

10. Type your first product code and key into the appropriate fields.

    The **Activator Wizard** is case-insensitive and displays only uppercase characters. It also ignores embedded spaces and separating characters. Product codes and keys do not contain the letters "O" or "I".

    > **CAUTION:** If you have a license for both the Windows runtime (`wrun32.exe`) and an Alternate Terminal Manager (ATM) runtime (`run32.exe`) for the same machine, be aware that the Activator Utility creates a license file named `wrun32.alc` for each of them. To avoid a situation in which the Activator Utility overwrites the license file for the second runtime:
    >
    > - Make a backup copy of the Windows run-time license file prior to creating (and renaming) the ATM run-time license.
    > - Create the ATM run-time license and rename it to match the executable (change `wrun32.alc` to `run32.alc`) before creating the Windows run-time license.

11. If you have more than one code and key pair to enter, select **More** after typing the first code/key pair. Each time you click **More** and add a new code and key, the **Activator Wizard** creates a separate license file for the product code and key you entered and returns you to the code and key entry screen. Repeat this process until you have entered all code and key pairs.
12. Click **Finish**, and then **Finish** again when the installation is complete.

> **Note:** If license activation was successful, but you get a message during product startup indicating that the license file cannot be found, the license file may not be in the correct directory. Products search for license files in the following order:
>
> 1. The default directory, which is determined by the License Activator by examining the value of the ALLUSERSPROFILE system environment variable, and adding the version and platform information to that location. For example:
>
> ```
> %ALLUSERSPROFILE%\Rocket Software\extend 11.0.0\x64\wrun32.alc
> ```

2. If the product cannot find the license file in the default location, it then looks in the directory containing the corresponding executable file; for example, the product searches for `wrun32.alc` in the directory containing `wrun32.exe`.

UNIX Installation

[Back to Installation Notes](#)

This section contains information on licenses for extend and installing this update on UNIX platforms.

Note: For additional information regarding UNIX installations including licensing and configuration options, see *UNIX Installation* in your ***extend*** *Interoperability Suite* documentation.

1. At a terminal prompt, change to the directory where you extracted the patch update files.
2. Run the installer as follows:

```
installer-name [-d installation-path]
```

   where *installer-name* is something similar to `setup_acucob1010pmk59shACU`.

3. Extract the EULA by running the installer again as follows:

```
installer-name -EULA [-d EULA-path]
```

   Use the `-d` option to extract the EULA to a directory other than the default, as specified by *EULA-path*.

4. After the installation is complete, test the installation as follows:
   a. Navigate to the `sample` sub-directory of your installation directory.
   b. Compile and run the `tour` program using the following commands:

```
ccbl tour.cbl
runcbl tour.acu
```

New Features in extend 11.0.0

This topic lists the new features in ***extend*** 11.0.0.

- Rocket® ACUCOBOL-GT® Extension for Visual Studio Code
- ACUCOBOL-GT
- AcuBench
- AcuSQL
- ACU-CBLFLAGS
- AcuToWeb

- AcuXDBC

Rocket® ACUCOBOL-GT® Extension for Visual Studio Code

The new ACUCOBOL-GT Extension for Visual Studio Code product enables you to create, edit, compile, and debug ACUCOBOL-GT code using Visual Studio Code software. The extension supports the importation of Rocket AcuBench projects, programs, and screens created using the ***extend*** AcuBench desktop product, including compiler settings. Once imported, you can continue improving the project in the extension by creating and editing project components, and generating and compiling code.

The documentation for this new feature comes in a separate publication, ACUCOBOL-GT Extension for Visual Studio Code.

ACUCOBOL-GT

New features for ACUCOBOL-GT include:

New scale layout manager
: The scale layout manager enables you to control every aspect of a window and its components. It is designed to assist in growing or shrinking a window in response to an NTF-SCALE scaling event. The window grows or shrinks when your end users scroll up or down with the mouse while holding down the **Ctrl** key. For details, see *The Scale Layout Manager*.

Grid controls
: This release provides several new features for Grid controls, including:

  - **Support for Check Box and Radio Button controls within Grid Cells**–Provides new properties that enable you to embed Check Box and Radio Button controls within Grid control cells. See *Controls Within Grid Cells* in your *ACUCOBOL-GT User's Guide* as a starting point for the details of this feature.
  - **Modify a row height programmatically**–The new ROW-HEIGHT special property enables you to modify a row height programmatically without resetting or repopulating the ROW-HEIGHTS property array. It also enables you to obtain the height of a specific row. See the *ROW-HEIGHT (numeric)* topic for details.
  - **Modify a column width programmatically**–The new COLUMN-WIDTH special property enables you to modify a column width programmatically without resetting or repopulating the COLUMN-WIDTHS property array. It also enables you to obtain the width of a specific column. See the *COLUMN-WIDTH (numeric)* topic for details.
  - **Manipulate the DISPLAY-COLUMNS property**–The DISPLAY-COLUMN special property enables direct, targeted manipulation of the DISPLAY-COLUMNS property. See the *DISPLAY-COLUMN (numeric)* topic for details.
  - **Specify text wrapping on a full row, column, or grid**–In previous releases, the WRAP-TEXT special property for the Grid control type enabled you to specify text wrapping in a single cell. In this release, WRAP-TEXT has been enhanced to enable you to specify text wrapping on a full row, full column, or a full grid. See *WRAP-TEXT (numeric)* in your *ACUCOBOL-GT User's Guide* for details.
  - **Control string truncation style**–The new COLUMN-USE-ELLIPSES property controls the string truncation style on a per-column basis. See *COLUMN-USE-ELLIPSES (numeric)* in your *ACUCOBOL-GT User Interface Programming* documentation for details.
  - **Smooth Grid scrolling**–The SMOOTH_SCROLL style for Grid controls enables smooth vertical and horizontal scrolling instead cell-based scrolling. See the *SMOOTH_SCROLL* entry in the style column of the Grid control *Common Properties* topic.
  - **Hide or reveal a hidden row in the user interface**–The ADJUSTABLE-ROWS style enables you to hide or reveal grid rows. See the *ADJUSTABLE-ROWS* entry in the style column of the Grid control *Common Properties* topic.
  - **Vertically center Grid row headings**–The CENTERED-ROW-HEADINGS style vertically centers grid row headings. See the *CENTERED-ROW-HEADINGS* entry in the style column of the Grid control *Common*

Properties topic.

- **Adjust the vertical alignment in a cell**–The VALIGNMENT special property enables you to adjust the vertical alignment of text within a targeted cell. See the *VALIGNMENT (alphanumeric)* topic for details.
- **Send an event for a changed row height**–The MSG-GRID-SHOW-ROW event is sent when a row's height changes. See the *MSG-GRID-SHOW-ROW (value 16448)* topic for details.
- **Send an event for a change in row visibility**–The MSG-ROW-HEIGHT-CHANGED event is sent when a row's visibility changes. See the *MSG-ROW-HEIGHT-CHANGED (value 16447)* topic for details.

Check Box, Push Button, and Radio Button controls

This release provides the following new features for Check Box, Push Button, and Radio Button controls:

- **Display a single bitmap image**–The new BITMAP-WIDTH special property along with the BITMAP-NUMBER special property enable the display of a single bitmap image from a bitmap strip alongside the text for Check Box, Push Button, and Radio Button controls. See *BITMAP-WIDTH (numeric)* for details.
- **Display without gray borders**–This feature provides a new NOBORDER style, which gives a more modern look, enabling buttons to fuse with the background of the user interface. See the *Common Properties* topics for the Check Box, Push Button, and Radio Button control types.

Wider Debugger window

In this release, you can make the run-time debugger window wider so that you can easily view code longer than 80 characters. See *Run-time Debugger* for details.

.NET 8 Support

In this release, support for .NET 8 has been added as follows:

- **Call .NET From COBOL**–Call .NET 8 assemblies from your ACUCOBOL-GT Windows applications using the new `--netcore` run-time option. See the *--netcore* topic in your ACUCOBOL-GT documentation for details.
- **Call COBOL from .NET 8**–The `--netdll` compiler option has been enhanced to generate wrapper assemblies compatible with both the .NET Framework and .NET 8. See the *--netdll* topic in your ACUCOBOL-GT documentation for details.
- **NETDEFGEN support for .NET 8**–The NETDEFGEN Utility, while continuing to support .NET Framework input assemblies, now also supports .NET 8 input assemblies. See *NETDEFGEN Utility Reference* for further information.

RMNet

RMNet for RESTful Web services has been updated with the following new features:

- **RMNet HttpDeleteWithPayload function**–This release provides support for RMNet HttpDelete with a payload via the HttpDeletePayload function. For details, see *HttpDeleteWithPayload* in your ACUCUBOL-GT documentation.
- **RMNet HttpPatch function**–This release provides support for RMNet HttpPatch function. This function serves as an API that enables execution of an HTTP PATCH request method for RESTful Web services. For details, see *HttpPatch* in your ACUCUBOL-GT documentation.

User-defined file systems

You can now define dynamically loaded file systems using the * AFSI_USER*x*_NAME and * AFSI_*xxxxx*_MODULE run-time environment variables. See *User-defined File Systems* for details.

AcuBench

Many AcuBench features are now available in a new extension for Visual Studio Code. See *ACUCOBOL-GT Extension for Visual Studio Code* for details.

AcuSQL

The AcuSQL pre-compiler has been updated to support a variety of additional SQL syntax, inline with newer SQL standards, including:

- **Analytic (window) functions**–the OVER clause and the PARTITION BY clause. Examples include LEAD, LAG and RANK.
- **String functions**–Examples include LEFT, CHARINDEX and CONCAT.
- **Sequence objects**–Enhancements for sequence objects, including support for the NEXT VALUE FOR statement.
- **Triggers**–Enhancements for triggers, including support for the ENABLE/DISABLE TRIGGER.
- **TRUNCATE TABLE**–Support for the TRUNCATE TABLE statement.

Use the newly supported syntax as part of valid SQL statements, and in accordance with the AcuSQL documentation. Also, ensure any syntax used is supported by the current version of your RDBMS.

For more information, see the *AcuSQL User's Guide*, and the *Supported Relational Database Management Systems (RDBMSs)* section in *extend System Requirements*.

ACU-CBLFLAGS

In this release, the ">>IMP" directive now takes two new options for ACU-CBLFLAGS. The PUSH and POP options enable you to adjust the data layout of library routines to ensure that the layout of data items passed from the code to a library routine match the alignment and size of the numeric data layout required by that library routine. See *The ">>IMP" Directive* for details.

AcuToWeb

The following new features are available in this release of AcuToWeb:

New framework for the AcuToWeb Gateway

In this release, we have replaced the Python-based AcuToWeb Gateway by a Spring Boot Java framework based Gateway, which:

- **Improves cross-platform compatibility**–The move to Java allows your product to work consistently on various operating systems and architectures, which can greatly expand your potential user base and customer reach.
- **Supports 32-bit and 64-bit platforms**–Java provides excellent support for both 32-bit and 64-bit platforms, allowing your compiled Java application to run on a wide range of systems without modification.
- **Enhances performance**–Java is known for its performance optimization and scalability.
- **Provides easier integration**–Java has a robust ecosystem of libraries, frameworks, and tools, making it easier to integrate your product with other systems, services, and technologies.
- **Ensures long-term viability**–Java is a well-established and widely used programming language, which means it is likely to be supported and maintained for many years to come. This can provide stability and longevity to your product.
- **Improves security and stability**–Java's security features and strong typing can help improve the security and reliability of your product, which is crucial for long-term success.
- **Enables dynamic CSS change**–Now, with the new Java framework, you can dynamically change the CSS you want to use for your applications without having to stop and restart the Gateway.

New AcuToWeb Browser Interface

This release provides a new AcuToWeb Browser Interface that replaces the UI for Connection Setup and the Themes Generator. It also provides a AcuToWeb live statistics with a summary of Connection Information. See *AcuToWeb Browser Interface* for details.

Updated Web site host capability

To better facilitate AcuToWeb as a Web Site Host, the `public_root_dir` property in the `gateway.toml` file now specifies a directory containing user Web files, and no longer represents the root directory that hosts the AcuToWeb Web files. See *Using AcuToWeb as a Web Site Host* in your *AcuToWeb User's Guide* for details.

AcuXDBC

AcuXDBC now supports running multiple queries in a single batch, each separated by a semicolon. During processing, if AcuXDBC encounters an error in a query statement, it ceases processing and does not execute subsequent queries in the batch.

> **Restriction:** The Batched Queries feature is not supported by command-line query tools such as the provided `asql.bat` script.

Known Issues

This topic describes issues that affect 11.0.0 applications if you are upgrading from a version of the ***extend*** Interoperability Suite earlier than version 11.0.0.

Back to top

> **Note:** See the *Known Issues and Restrictions* topic in the *Product Information* section of the ***extend*** Interoperability Suite publication for additional information about known issues.

The following are known to be issues that might affect existing applications created in a version of ***extend*** earlier than 11.0.0.

- The WIN$PRINTER opcode WINPRINT-SETUP-OLD has been removed. This was used in previous versions for compatibility with 16-bit Windows, which is no longer supported.
- The **vio** (`vio32.exe`) utility has been removed from version 11.0.0. If you need to use **vio** for any reason, keep a version from a previous release; however, we suggest that you use an alternative to **vio** when possible.
- The **sub** interface (also called the **C-74** interface for calling C routines) has been removed from version 11.0.0. If your applications call C routines using this interface, either update to use a later interface such as **sub85** or **direct**, or create bridge functions.
- Due to rebranding from Micro Focus® to Rocket Software, our .NET interface MICROFOCUS.NETOBJECT stored object class has been renamed to ROCKETSOFTWARE.NETOBJECT. Either regenerate your .NET copybooks using the 11.0.0 NETDEFGEN utility (preferred), or manually edit your existing copybooks to use the new class name.
- The compiler no longer supports the creation of object code for run-time versions 8.0 and earlier. If you regularly use the `-Z`*nn* option to generate object code for use with the version 8.0 or earlier runtime, we recommend that you upgrade your run-time version, or update your application to work around this issue.

Enhancements

This section lists issues that have been resolved by providing a product enhancement to this release.

Back to top

> **Note:** The numbers that follow each issue listed in this *Enhancements* section represent the Support Case Number followed by the issue number in parentheses.

- Acu4GL
- ACUCOBOL-GT
- AcuXDBC

Acu4GL

Back to top of Enhancements

- MSSQL: A new configuration variable, A_MSSQL_IDENTITY_TYPE, has been added to determine the type of identity column created when a table is created using Acu4GL for SQL Server. This enables you to store a 64-bit value internally, which can prevent overflows caused by 32-bit integers. For details, see *A_MSSQL_IDENTITY_TYPE* in your Acu4GL documentation.

  02341569 (296069)

ACUCOBOL-GT

Back to top of Enhancements

- `rmnet.dll`: This release provides a new run-time configuration file, MAX_DUMP_LINES, that you can use with ACU_DUMP to prevent a dump file from growing too large over time and thus using up excessive disk space. For details, see *MAX_DUMP_LINES* in your *ACUCOBOL-GT Reference Manual*.

  02325873 (286187)

- Runtime: This release provides a new configuration option, A_FLOAT_ROUND_METHOD, that enables you to change the method used when rounding floating-point numbers. The default method converts to fixed-point with rounding using a maximum number of digits after the decimal point. This can sometimes produce an unexpected result. The new method converts to fixed-point with rounding using a limited number of digits after the decimal point. In some cases, this produces a more precise result. For details, see *A_FLOAT_ROUND_METHOD* in your *ACUCOBOL-GT Reference Manual*.

  02660555 (544074)

- Compiler: In previous releases, ACU support for connecting to the WebSphere MQ interface through the USE_MQSERIES configuration variable was limited to Windows 32-bit only. This enhancement extends our support to include Windows 64-bit and Linux. For details about the USE_MQSERIES configuration variable, see *USE_MQSERIES* in your ACUCUBOL-GT documentation.

  02608402 (506136)

- As disk space is now rarely an issue, the **`vutil -rebuild`** option has been modified to remove the `-s` option, including its `-r`, and `-m #` options. These options allowed a rebuild to use removable disks to hold the records when local storage was insufficient to hold the original file and the new file during the rebuild.

  (658076)

- Valid values for the WMENU-POPUP parameter of the W$MENU library routine now include negative numbers. For complete details, see the *W$MENU* library routine reference topic in your ACUCOBOL-GT documentation.

  00365213 (11535)

- The Sub interface, also known as the C-74 interface, has been removed from ACUCUBOL-GT as an option for calling C functions from COBOL due to its obsolescence and inefficiency.

- The DEFER_EVENTS configuration variable has been updated to fix a problem with recursive behavior that sometimes occurred when an event procedure was long running. A third option, `2` or `QUEUE`, has been added to address this issue. For details, see *DEFER_EVENTS* in your product Help.

  In addition, this ECN also fixes an issue with the debugger that caused the command window cursor to flash under certain conditions.

  02911389 (738077)

- The Vision File System Interface has been updated to use modern features.

AcuXDBC

Back to top of Enhancements

> **Attention:** If you are upgrading from a previous release of AcuXDBC, before using AcuXDBC 11.0, we recommend that you reload all XFDs into the system catalog to enable certain enhancements. If reloading all XFDs is not possible, run `xdbcutil -g 2` on your system catalog before starting AcuXDBC. For details, see *xdbcutil utility* in your *AcuXDBC User's Guide*.

- The `xdbcutil` command has been updated to support two new virtual columns into which you can load relative XFDs. See *xdbcutil utility* and *System Catalog Extension for Relative Files* in your *AcuXDBC User's Guide* for details.

  02629762 (529079)

- Compiler/Runtime: The new CARRIAGE-CONTROL-FILTER XFD directive provides the same functionality as the CARRIAGE_CONTROL_FILTER run-time configuration variable, allowing carriage control characters to be retained when AcuXDBC reads a LINE SEQUENTIAL data file. See *CARRIAGE-CONTROL-FILTER directive* in your *AcuXDBC* documentation for details.

  02669873 (557005)

- You can now run multiple queries in a single batch, each separated by a semicolon. If a statement contains an error, all previous statements in the batch are executed, but subsequent statements are not executed.

  > **Note:** Batched queries are not supported within command-line query tools such as the provided `asql.bat` script.

  02667872 (542100)

- SELECT query times have been reduced by as much as 50% for columns of a COMP-3 type with Signed packed-decimal (field-type 9 in the XFD file) and Positive packed-decimal (field-type 8) data.

  02837524 (672051)

Resolved Issues

This section lists issues that have been resolved by providing a product enhancement to this release.

Back to top

Note: The numbers that follow each issue listed in this *Resolved Issues* section represent the Support Case Number followed by the issue number in parentheses.

- Acu4GL
- AcuBench
- ACUCOBOL-GT
- AcuConnect
- AcuSQL
- AcuToWeb
- AcuXDBC

Acu4GL

Back to top of Resolved Issues

- When using AcuXDBC or Acu4GL, records did not always respect XFD WHEN conditions specified in the FD of the record, specifically when using AND and OR conditions. This has been fixed.

  02842010 (673025)

- Programs that used CALL SQL.ACU with an SQL command longer than 2000 characters caused the statements that access tables to generate random errors. This has been fixed to accommodate SQL commands up to 50,000 characters.

  03052419 (ACGL-73)

AcuBench

Back to top of Resolved Issues

- Multiple bitmaps included in a single report sometimes caused the source indentation for a generated END-IF to become negative, resulting in the generation of invalid source code. This has been corrected.

  (542069)

- A problem that caused AcuBench to freeze when modifying the Grid control NUM-ROWS property has been fixed.

  (ACUW-2683)

- The FD files were missing from the AcuBench `Samplprj` sample project, causing AcuBench to crash during generation. The files are now included in the sample project.

  (ACUW-2650)

- A problem that caused AcuBench to crash on copyfile generation has been fixed.

  (ACUW-2681)

- The layout manager was erroneously duplicating values in the Working Storage when a graphical control was cut and then pasted. This has now been fixed.

(ACUW-2675)

ACUCOBOL-GT

Back to top of Resolved Issues

- When converting between string tables and string arrays using C$JAVA with the CJAVA-CONVERTTABLETOARRAY and CJAVA-CONVERTARRAYTOTABLE operations, the character set was not converted per the A_JAVA_CHARSET setting. This sometimes resulted in string data being converted to UTF-8 by Java and not being converted back to the original character set. This problem has been corrected.

  (534002)

- Repeatedly calling Java using C$JAVA caused an error. A fix is available in the 10.5.1 release, and has been updated in this release to improve performance in threaded COBOL programs.

  00371299 (306036)

- The V104_BITMAPS configuration variable has been updated to behave as documented, meaning that value 1 is equivalent to BUTTON, 2 is equivalent to CONTROL, 4 is equivalent to IMAGELIST, 8 is equivalent to DISPLAY, and 16 is equivalent to PRINT. In addition, a problem that prevented the value V104_BITMAPS from being transmitted to Thin Client has been fixed.

  02729546 (629005, 629033)

- Text wrapped over multiple lines always left-aligned, even when center- or right-alignment was specified. This is now resolved; text aligns correctly with wrap-text, provided the cell is large enough to display the text.

  (572012)

- When a COBOL program used C$LIST-DIRECTORY to access multiple directories on the Thin Client host, the runtime sometimes crashed when trying to get files from any directory except the first. This has been corrected.

  (649005)

- Performing many ESQL calls in an object compiled for native code sometimes caused an unintended recursive call loop, which eventually overflowed the stack and caused a memory error. This has been corrected by implementing a change to the way the runtime handles the return of ESQL calls when compiled for native code.

  02757428 (640133)

- A memory access violation occurred when displaying a grid containing 0 rows. This has been fixed.

  02744155 (632008)

- The runtime intermittently threw a memory access violation during user keyboard input when all grid rows were column headers. This has now been fixed.

  02779399 (644025)

- The runtime sometimes threw a memory access violation (MAV) when a grid was reset. This has been fixed.

  02754939 (637063)

- Windows using the layout manager could not be resized to fit the full width of the monitor when resized by dragging the edge of the window. This is now fixed with the window now properly resizing to the edge.

  02767669 (637127)

- When using both the PAGED style and the VSCROLL style, the paged grid buttons (used for moving down a line, down a page, etc.) did not respond to clicks. This has been corrected.

  02804141 (651141)

- This change corrects the behavior of JSON GENERATE SUPPRESS WHEN to allow use on OCCURS items.

  02726415 (640096)

- Doing a DISPLAY SCREEN with a grid control sometimes caused a memory access violation. This has been fixed.

  02786675 (656065)

- When used with Thin Client, the debugger window, and potentially other windows as well, resized itself when moved. This has been fixed.

  (653218)

- In a grid using multi-row records and column headings, the cursor sometimes appeared over the incorrect cell. This is now resolved.

  (653233)

- A problem that prevented scrolling to the bottom row of grids that had ROW-DIVIDERS = 0 set has been fixed.

  02840341 (672077)

- When bitmaps in grid cells used a transparent background, they could display incorrectly. This has been fixed.

  02729546 (629003)

- CENTER- and RIGHT-aligned text within a grid cell also containing a bitmap displayed as misaligned. This has been fixed.

  02741935 (627111)

- In grids reloaded cell by cell, a previously sorted column could not be resorted when the resort followed a RESET-GRID command. This has been fixed.

  > **Note:** Grids reloaded record by record were not affected.

  02754939 (589014)

- A timing problem with the Thin Client sometimes caused a memory access violation when a user interaction was executed while the runtime was waiting for a response from the Thin Client. This has been fixed.

  02759686 (640129)

- This release contains a fix for a problem that caused a grid to crash when it contained a large number of rows (for example, more than 254) and when NUM-ROWS was set to -1.

02890645 (689091)

- When using the Thin Client to debug a remote process using the --wait option, the remote process either never connected to the Thin Client, or the connection took an excessive amount of time. This has been corrected.

  02906407, 02932091 (703012)

- A problem that caused the scroll bar to appear with an incorrect range when a Grid control HSCROLL or VSCROLL style was set after grid initialization (i.e., not in the SCREEN SECTION) has been fixed.

  (725050)

- An `Error: EmptyNativeFunction() called!` sometimes occurred when executing native code objects with the AIX 7.3 port. This has been fixed.

  02958853 (732006)

- The DATA-RANGE and EXCEPTION-RANGE commands of the KEYBOARD configuration variable had no effect when running under Thin Client. This has been fixed.

  02935318 (712119)

- When processing events on an INQUIRE, Thin Client sometimes failed to wait for one process to complete before executing the next process. This caused problems when the currently processing event failed. This problem has been corrected.

  02935318 (712119)

- An error that sometimes occurred when using the `-Lx` option caused the Descriptor Words value in an extended listing to be incorrect. This has been fixed.

  02795477, 01022917 (651085)

- The transparent background of a bitmap on a push button displayed incorrectly on a subsequent window when the bitmap was first used on a previous window. This has been corrected.

  01018387, 01019337 (ACUBR-296)

- Setting BITMAP to NULL failed to remove the image from a Grid control. This is now fixed.

  01023029 (704029)

- An ACCEPT FROM ENVIRONMENT statement in a large COBOL program sometimes caused a memory violation error due to use of a data descriptor over 32,767. Data descriptors are internal compiler/runtime indexes used to identify individual data items. This problem has been fixed.

  01022941 (637073)

- In very large programs, data descriptors greater than 0x7FFF sometimes caused a `reference modification descriptor too large` error. This issue has been fixed.

  01004706, 01022917 (542017, 651085)

- A problem that caused combo boxes to show too few items in the drop-down list when using the WIN32_NATIVECTLS configuration variable and when the height of the combo box was specified in pixels has been fixed. A problem that caused a combo box with its height specified in pixels to have a drop-down list that spilled beyond the bounds of the main window has also been fixed.

01022791 (748083)

- A problem that caused the precompiler to fail when compiling large COBOL programs with Boomerang has been fixed.

  03058241 (ACUB-2637)

- When inserting directives into a preprocessed file where the original file had an EXEC SQL ... END-EXEC statement as the last line of the file, Boomerang would leave this EXEC statement at the end of the file instead of omitting it.

  01095127 (ACUB-2763)

- The runtime sometimes failed to accept input from an Activex DATE control. This has been fixed.

  01098924 (ACUB-2787)

- Screen images sometimes failed to display after a window was minimized and then restored. This has been fixed.

  01099287 (ACUB-2789)

- A problem that sometimes occurred when one program called another resulted in the called program return to the calling program at an incorrect point in the code. This has now been corrected.

  01099500 (ACUB-2792)

- Using the Page Up and Page Down features on a non-paged Grid resulted in some records being skipped. This has been corrected.

  01090954 (ACUB-2692)

- The compiler sometimes threw an `Out of Main Memory!` error, resulting in a defective `.acu` file. This has been fixed.

  01098862 (ACUB-2786)

- A problem that sometimes caused the compiler throw a `realloc(): invalid pointer` error has been fixed.

  SC-01105625 (ACUB-2969)

AcuConnect

[Back to top of Resolved Issues]

- When starting AcuThin, the splash screen failed to show. This has been fixed.

  02863874 (685130)

AcuSQL

[Back to top of Resolved Issues]

- Some clauses, such as LIKE, that used a COBOL data-item parameter containing the % wildcard character sometimes incorrectly returned `SQL ERROR NO DATA (code 100)`. This has been corrected.

  > **Important:** For some clauses, the trailing spaces of a data item are significant. As such, for the SQL DBMS to return expected output, it might be necessary to strip trailing spaces and replace them with low values to effectively ensure that the string is null-terminated.

  02461277 (392021)

- Size information was lost on wide CHAR parameters, generating either of the following errors:

  - Conversion to UNICODE using the specified character set not taking place when performing an INSERT.
  - Server returning SQL_NO_DATA (error code 100) when using a parameter bound to an N(VAR)CHAR column in a WHERE clause.

  These have been fixed.

  02610856, 02781932 (509038, 643044)

- For a given statement, subsequent INSERTs and UPDATEs of a Unicode (NACHAR or NVARCHAR) column sometimes reinserted the value of the statement's first INSERT or UPDATE. This has been fixed, ensuring that the database receives the expected data from its parameters after the first INSERT or UPDATE.

  02820452 (662043)

- A problem that prevented the SQL error code from being set following an update of a non-existent record has been fixed such that the SQLCA is now updated with the expected error code when the statement execution returns SQL_NO_DATA (value 100).

  02820452 (662043)

- Certain statements operated on a connection other than the SET connection when multiple connections were open. To fix this issue, the SQL SET CONNECTION statement has been updated to always apply subsequent queries to the currently set connection.

  02804235 (658053)

AcuToWeb

[Back to top of Resolved Issues]

- A problem that caused entry fields on a mobile device to behave incorrectly has been fixed.

  03045155 (ACUW-2407)

- A problem that occurred when opening a new COBOL window in front of a window that contained a browser control caused the browser control to reload the Web page. This has been corrected.

  02909862 (1533055)

- A problem that sometimes prevented a combo-box control from closing after clicking away from the control has now been fixed.

  02982735 (1585026)

- The AcuToWeb TreeView sometimes failed to properly trigger click events. This has been fixed.

  01023104 (ACUW-2474)

- A problem that prevented the correct selection of TreeView elements using the keyboard has been fixed.

  01084927 (ACUW-2484)

- A MSG-SB-NEXT event was sometimes erroneously triggered when scrolling upward using SCROLL-BAR upward. This could also cause the scrolling to flicker back and forth, making it difficult to navigate back to the first row. This problem has been fixed.

  01082829 (ACUW-2471)

- AcuToWeb sometimes threw an error when it encountered a relative path in COBOL source. This has been corrected.

  01099378 (ACUW-2672)

- AcuToWeb failed to generate certain events for applications using the Tab-Control with BUTTONS style. This has been fixed.

  01104048 (ACUW-2734)

- Push Buttons were sometimes rendered incorrectly due to a problem with the WIDTH property value. This has now been corrected.

  01102835 (ACUW-2725)

AcuXDBC

Back to top of Resolved Issues

- Compiler: INSERT statements containing a SELECT sub-query are now handled correctly. For example, assuming TABLE1 and TABLE2 are sufficiently compatible, the following statement correctly inserts two rows from TABLE2 into TABLE1:

  ```
  INSERT INTO TABLE1 SELECT TOP 2 SKIP 1 * FROM TABLE2;
  ```

  02694745 (593005)

- Inserting data into an element of a large array (OCCURS structure in COBOL) sometimes caused the inserted data to be truncated. This has been fixed.

  > **Attention:** To use this fix, either reload all XFDs into the system catalog, or run `xdbcutil -g 2` on your system catalog before starting AcuXDBC. For details, see *xdbcutil utility* in your *AcuXDBC User's Guide*.

  02712623 (614079)

- Binary data was sometimes incorrectly interpreted as a hex string on an INSERT into a field marked as binary in a line-sequential file. This has been fixed, with the expected data correctly written to the corresponding field in the line-sequential file.

02805698 (656058)

- When comparing fields against spaces in a WHERE clause, AcuXDBC incorrectly required that number of spaces match the size of the field. This has been fixed such that a single space in quotes (`' '`) is enough for the comparison to give expected results.

  02824579 (661074)

- To improve performance when processing multiple queries made within a short interval of time, support for the Windows-platform QUERY_TIMEOUT configuration variable has been removed.

  02796888 (653097)

- A problem with the XFD SUBTABLE directive caused a partial add of alternate record keys to a subtable. This has been fixed.

  > **Note:** As documented, alternate keys do not appear as columns in the subtable; only primary keys appear.

  02820697 (662051)

- A problem that caused line-sequential file WRITE performance to degrade as the file grew larger has been fixed.

  01018909 (ACX-106)

- A problem that caused binary sequential files to hang on an INSERT has been fixed.

  01020171 (ACX-107)

- A catalog load error occurred when creating the AcuXDBC system catalog if a view was created using the XFD WHEN directive with the TABLENAME clause, and that same view also included an XFD WHEN directive for subordinate fields. For example, an XFD WHEN directive for subordinate fields might be added to handle a REDEFINES clause. This is now fixed. The TABLENAME clause is now compatible with XFD WHEN directives applied to subordinate fields.

  01022929 (ACUB-2759)

Contacting Rocket Software

Our Web site gives up-to-date details of contact numbers and addresses.

Additional technical information or advice is available from several sources.

To view the support pages, enter https://www.rocketsoftware.com/ in your browser to go to the Rocket Software home page, then click **Support**. Select one of the available support options such as **Submit Case**.

If you are a customer of Technical Support, please see the *Support Handbook* that includes information about downloading and licensing your product, contacting Technical Support, and about reporting an incident. You can download it from our Web site at https://www3.rocketsoftware.com/support-manual/pdf/Rocket_Software_Technical_Support_Users_Guide_current.pdf. Support from Rocket Software may be available only to customers who have maintenance agreements.

You can download fixes and documentation updates as follows:

1. Log into the Rocket Customer Community site at https://my.rocketsoftware.com/RocketCommunity/s/.
2. Click **Downloads**.
3. Locate and select the product to download in the **SELECT A PRODUCT** pane.
4. Click on the link in the center pane for the product you want to download.
5. Click the required file link in the **File Name column** on the relevant row.

Trademark and Legal Information

This topic contains trademark and legal information pertaining to this Rocket Software product.

Rocket® ***extend*** Interoperability Suite (formerly a product of Micro Focus or OpenText) may contain outdated references to "Micro Focus" or "OpenText", which are trademarks of OpenText. Rocket Software is not affiliated with Micro Focus or OpenText and has rebranded these products as Rocket® products.

Rocket® Trademarks

The Rocket® extend™ Interoperability Suite contains the following products:

- Rocket® Acu4GL® for DB2
- Rocket® Acu4GL® for Informix
- Rocket® Acu4GL® for MS SQL
- Rocket® Acu4GL® for ODBC
- Rocket® Acu4GL® for Oracle
- Rocket® AcuBench™
- Rocket® ACUCOBOL-GT® Dev Suite
- Rocket® ACUCOBOL-GT® Dev Sys
- Rocket® ACUCOBOL-GT® Runtime (CGI)
- Rocket® ACUCOBOL-GT® Runtime
- Rocket® AcuConnect® - Distributed Processing
- Rocket® AcuConnect® - Thin Client
- Rocket® AcuServer™
- Rocket® AcuSQL™ Precompiler
- Rocket® AcuSQL™ Runtime
- Rocket® AcuToWeb™
- Rocket® AcuXDBC™
- Rocket® AcuXDBC™ Enterprise Edition
- Rocket® AcuXDBC™ Server
- Rocket® Dedicated Master Runtime
- Rocket® Xcentrisity Business Information Server for extend™

IBM® Trademarks

The ***extend*** Interoperability Suite references IBM® COBOL, IBM® CICS®, IBM® IMS™, IBM® Db2®, and IBM® WebSphere MQ®, all of which are trademarks of International Business Machines Corporation, registered in many jurisdictions worldwide.

End User License Agreement (EULA)

You can view the EULA from the Rocket® Agreements Web page.

Creative Commons License

Portions of some programming examples were produced by Dr. Frank McCown, Harding University Computer Science Dept and are licensed under a Creative Commons License. Such portions have been modified by Rocket Software.